



VIVEKANAND EDUCATION SOCIETY'S

Institute of Technology

(An Autonomous Institute Affiliated to University of Mumbai, Approved by A.I.C.T.E & Recognized by Govt. of Maharashtra)

Department of Artificial Intelligence and Data Science

Artificial Intelligence Theory

Course Code	Course Name	Teaching Scheme (Teaching Hours)			Credits Assigned			
		Theory	Practical	Tutorial	Theory	TW/PR	Tut	Total
NADPC43	Artificial Intelligence (Theory)	03	---	---	03	---	---	03

Cours e Code	Cours e Name	Examination Scheme					
		Theory			Term Work	Practica l & Oral	Total
		Internal Assessment		End Sem Exa m			
		Mid- Term Test	Continuous Assessment				
NADPC43	Artificial Intelligenc e (Theory)	20	20	60	---	---	100

Course Prerequisite:	
Course Objectives:	
1	To gain perspective of AI and its foundations.
2	To study different agent architectures and properties of the environment
3	To understand the basic principles of AI towards problem solving, inference, perception, Knowledge representation, and learning.
4	To investigate probabilistic reasoning under uncertain and incomplete information.
5	To explore the current scope, potential, limitations, and implications of intelligent systems
Course Outcomes:	
After successful completion of the course students will be able to:	
1	Identify the characteristics of the environment and differentiate between various agent architectures.
2	Apply the most suitable search strategy to design problem solving agents.
3	Represent a natural language description of statements in logic and apply the inference rules to design Knowledge Based agents.
4	Apply a probabilistic model for reasoning under uncertainty.
5	Comprehend various learning techniques.
6	Describe the various building blocks of an expert system for a given real word problem.



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Course Code	Course Name	Teaching Scheme (Teaching Hours)			Credits Assigned			
		Theory	Practical	Tutorial	Theory	Practical	Tut	Total
NADPC43	Artificial Intelligence (Lab)	---	02	---	---	01	---	01

Cours e Code	Course Name	Examination Scheme					
		Theory			Term Work	Practica l & Oral	Total
		Internal Assessment		End Sem Exa m			
		Mid- Ter m Test	Continuous Assessment				
NADPC 43	Artificial Intelligence (Lab)	---	---	---	25	25	50

Lab Prerequisite:	
Lab Objectives:	
1	To design suitable Agent Architecture for a given real world AI problem
2	To implement knowledge representation and reasoning in AI language
3	To design a Problem-Solving Agent
4	To incorporate reasoning under uncertainty for an AI agent
Lab Outcomes:	
After successful completion of the course students will be able to:	
1	Identify suitable Agent Architecture for a given real world AI problem
2	Implement simple programs using Prolog
3	Implement various search techniques for a Problem-Solving Agent
4	Represent natural language description as statements in Logic and apply inference rules to it
5	Construct a Bayesian Belief Network for a given problem and draw probabilistic inferences from it

Artificial Intelligence (Theory)

Mod u le		Detailed Content	Hours
1		Introduction to Artificial Intelligence	3
	1.1	AI, The Foundations of Artificial Intelligence ,The History of Artificial Intelligence, The State of the Art, Risks and Benefits of AI	
	1.2	Agents and Environments, Good Behavior: The Concept of Rationality, The Nature of Environments, The Structure of Agents	
2		Problem-solving	4
	2.1	Solving Problems by Searching : Problem-Solving Agents, Example Problems, Search Algorithms, Uninformed Search Strategies, Informed (Heuristic) Search Strategies, Heuristic Functions.	
	2.2	Search in Complex Environments: Local Search and Optimization Problems, Local Search in Continuous Spaces, Search with Nondeterministic Actions, Search in Partially Observable Environments, Online Search Agents and Unknown Environments .	
	2.3	Adversarial Search and Games: Game Theory, Optimal Decisions in Games, Heuristic Alpha–Beta Tree Search, Monte Carlo Tree Search, Stochastic Games, Partially Observable Games, Limitations of Game Search Algorithms.	
3		Knowledge, reasoning, and planning	12
	3.1	Logical Agents: Knowledge-Based Agents, Overview of Propositional Logic.	
	3.2	First-Order Logic: Representation Revisited, Syntax and Semantics of First-Order Logic, Using First-Order Logic, Forward Reasoning, Backward Reasoning, Resolution in First Order Logic.	
	3.3	Knowledge Representation: Ontological Engineering, Categories and Objects, Events, Mental Objects and Model Logic, Reasoning Systems for Categories, Reasoning with Default Information.	
	3.4	Automated Planning: Definition of Classical Planning, Algorithms for Classical Planning, Heuristics for Planning, Hierarchical Planning.	
4		Uncertain knowledge and reasoning	
	4.1	Quantifying Uncertainty: Acting under Uncertainty, Basic Probability Notation, Inference Using Full Joint Distributions, Independence, Bayes' Rule and Its Use, Naive Bayes Models.	
	4.2	Probabilistic Reasoning: Representing Knowledge in an Uncertain Domain, The Semantics of Bayesian Networks, Exact Inference in Bayesian Networks.	
	4.3	Probabilistic Reasoning over Time: Time and Uncertainty, Inference in Temporal Models, Hidden Markov Models.	
5		Natural Language Processing	
	5.1	Introduction: Origin & History of NLP, The need of NLP, Generic NLP System, Levels of NLP, Knowledge in Language Processing,	



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		Ambiguity in Natural Language, Challenges of NLP, Applications of NLP.	
	5.2	Applications of NLP, Overview of Machine Learning and Deep Learning	
6		Modern AI Applications	
	6.1	Generative AI: What is Generative AI?, Generative AI vs. Traditional AI, How Does Generative AI Work?, Types of Generative AI, Real-world Uses for Generative AI.	
	6.2	Reinforcement Learning: Introduction, Examples, Elements of Reinforcement Learning, Limitations and Scope, An Extended Example: Tic-Tac-Toe, Early History of Reinforcement Learning .	
	6.3	Ethics in AI : What are Ethics?, Types of Ethics, Ethical issues in Artificial Intelligence.	

Textbooks:	
1	Stuart J. Russell and Peter Norvig, "Artificial Intelligence A Modern Approach —Second Edition" Pearson Education.
2	Elaine Rich and Kevin Knight —Artificial Intelligence Third Edition, Tata McGraw-Hill Education Pvt. Ltd., 2008.
3	George F Luger —Artificial Intelligence Low Price Edition, Pearson Education., Fourth edition.
References:	
1	Ivan Bratko —PROLOG Programming for Artificial Intelligence, Pearson Education, Third Edition.
2	D. W. Patterson, Artificial Intelligence and Expert Systems, Prentice Hall.
3	Saroj Kaushik —Artificial Intelligence, Cengage Learning.
4	Davis E. Goldberg, —Genetic Algorithms: Search, Optimization and Machine Learning, Addison Wesley, N.Y., 1989.
5	Patrick Henry Winston, —Artificial Intelligence, Addison-Wesley, Third Edition.
6	N. P. Padhy, —Artificial Intelligence and Intelligent Systems, Oxford University Press.

Access to software and virtual labs:

<https://www.swi-prolog.org/download/stable>

Industry articles and case studies :

<https://www.digitaleurope.org/news/case-studies-on-artificial-intelligence/>

<https://www.linkedin.com/pulse/rise-artificial-intelligence-comprehensive-case-flytc/>

Any other (Access to AI tools / Data driven insights (if applicable) or any other):

<https://www.kaggle.com/>



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Internal Assessment:

Assessment consists of one Mid Term Test of 20 marks and Continuous Assessment of 20 marks. Mid Term test is to be conducted when approximately 50% syllabus is completed. Duration of the midterm test shall be one hour.

Continuous Assessment:

Continuous Assessment is of 20 marks. The rubrics for assessment will be considered based on approval by the subject teachers. The rubrics can be any 2 or max 4 of the following:

Sr. No.	Rubrics	Marks
1	*Certificate course for 4 weeks or more: NPTEL/ Coursera/ Udemy/any MOOC	10 marks
2	Content beyond syllabus presentation	10 marks
3	Creating Proof of concept	10 marks
4	Mini Project / Extra Experiments/ Virtual Lab / Competitive programming-based event / Group Discussion	10 marks
5	Multiple Choice Questions (Quiz)	5 marks
6	GATE Based Assignment/Tutorials etc	10 marks

*For sr. no. 1, the date of the certification exam should be within the term and in case a student is unable to complete the certification, the grading has to be done accordingly.

End Semester Theory Examination:

1	Question paper will be of 60 marks
2	Question paper will have a total of five questions
3	All questions have equal weightage and carry 20 marks each
4	Any three questions out of five needs to be solved.

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Artificial Intelligence (Lab)

Suggested Experiments:	
Sr. No.	Name of the Experiment
1	Provide the PEAS description and TASK Environment for a given AI problem , Identify suitable Agent Architecture for the problem
2	Formulate the Problem Statement for the AI System
3	Implement any one Uninformed Search (DFS/BFS) for 4 queen/8 queen/8 puzzle or any state space
4	Implement any one of the Informed search techniques E.g. A-Star algorithm for 8 puzzle problem/ TSP
5	Implement any one of the Local Search techniques. E.g. Hill Climbing, Simulated Annealing, Genetic algorithm for 4 queen
6	Prove the goal sentence from the following set of statements in FOPL by applying forward, backward and resolution inference algorithms. wampus/ resolution graph
7	Prove the goal sentence from the following set of statements in FOPL by applying forward, backward and resolution inference algorithms. wampus/ resolution graph
8	Case study of any existing successful AI system

Note: Suggested List of Experiments is indicative. However, flexibilities lie with individual course instructor to design and introduce new, innovative and challenging experiments, (limited to maximum 30% variation to the suggested list) from within the curriculum, so that, the fundamentals and applications can be explored to give greater clarity to the students and they can be motivated to think differently.

Term Work:	
1	Term work should consist of 10 experiments.



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2	The final certification and acceptance of term work ensures satisfactory performance of laboratory work and minimum passing marks in term work.
3	Total 25 Marks (Experiments: 15-marks, Term work Assessment: 10-marks)

Evaluation Exam	
1	Practical Exam based on the entire syllabus